

Experiment No : 05

Experiment Name : Study of single-stage BJT amplifier circuit

Objectives:

- To implement a single-stage BJT amplifier circuit
- To observe the input and output waveforms of a BJT amplifier circuit

Theory:

A single-stage BJT amplifier is used to amplify weak signals with voltage gain. The transistor operates in active region, with a voltage divider biasing network ensuring a stable operating point. An AC coupling capacitor blocks DC components from the input signal.

Required Apparatus:

- BJT (BD135, 1 piece)
- Resistors ($1\text{ k}\Omega$, $2.2\text{ k}\Omega$, $4.7\text{ k}\Omega$: 1 piece each)
- Potentiometer ($100\text{ k}\Omega$, $50\text{ k}\Omega$: 1 piece each)
- DC power supply (0-15V)
- Project board
- Multimeter
- Connecting wires

Circuit Diagram:

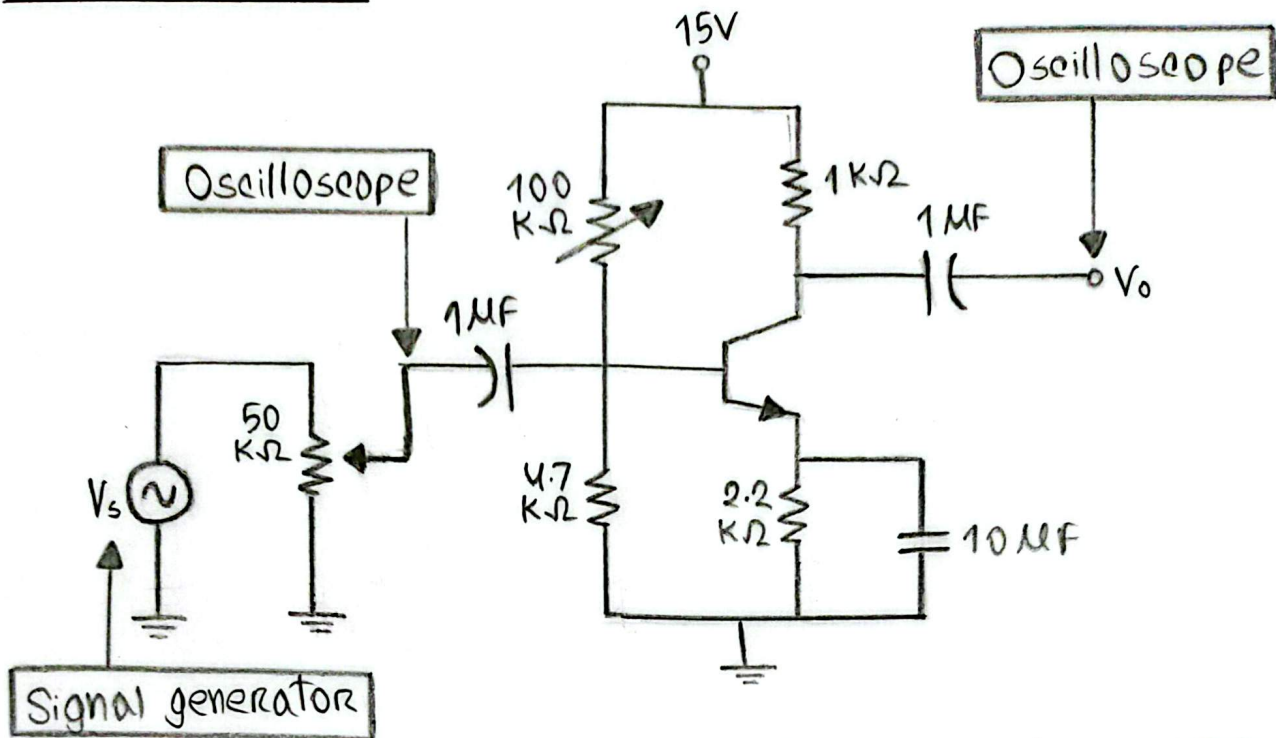


Figure: Circuit Diagram for implementing single-stage BJT amplifier

Procedure:

- (i) The circuit was connected according to the circuit diagram without the capacitors
- (ii) The mid point biasing ($V_{CE} = \frac{V_{CC}}{2}$) was performed by tuning the potentiometer
- (iii) After the completion of biasing, connect the coupling capacitors and bypass capacitors
- (iv) A small signal to the input side was supplied using signal generator and the corresponding output waveform in the oscilloscope was recorded
- (v) Lastly, the experimental and theoretical voltage gain of the amplifier circuit were compared

Calculations:

$$\begin{aligned}\text{Theoretical gain, } A_v &= -\frac{R_c}{R_E} \\ &= -\frac{1000}{220} \\ &= 4.5454\end{aligned}$$

$$\begin{aligned}\text{Measured gain, } A_v' &= \frac{V_o(p-p)}{V_i(p-p)} \\ &= 4.8/1.04 \\ &= 4.615\end{aligned}$$

$$\begin{aligned}\therefore \text{Error} &= \frac{|A_v - A_v'|}{A_v} \times 100\% \\ &= \frac{|4.5454 - 4.615|}{4.5454} \times 100\% = 1.55\%\end{aligned}$$

Discussion and Conclusion:

In this experiment, a single-stage BJT amplifier was implemented and proper signal amplification was observed. The measured gain was close to the theoretical value, achieving the objectives of the experiment.

Precautions:

- All the connections should be correct
- Transistor terminals must be identified properly
- Calibration must be performed properly